Check for updates

Original Article

QJEP

Quarterly Journal of Experimental Psychology
1–20
© Experimental Psychology Society 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/17470218261429045
qjep.sagepub.com
Sage

From Features to Identity: Evidence for a Hierarchical Model of Unfamiliar Face Matching

Daniel Fitousi[1]

Abstract

How does nominal identity emerge from featural information in a face-matching task with unfamiliar faces? The present study extends the classic same–different paradigm to the domain of unfamiliar face matching and proposes a hierarchical serial self-terminating model. The model assumes that similarity between images of faces can be assessed at three representational levels—featural, physical, and nominal—and that observers compare faces sequentially from lower to higher levels, guided by instructional set (physical vs. nominal) and terminating once a discriminating feature is found. Two experiments tested these predictions with upright (Experiment 1) and inverted (Experiment 2) faces. Upright faces showed key diagnostic effects consistent with the model: the absence of a fast-same effect, the presence of the name–physical disparity effect, and costs of nominal relative to physical instructions. Signal-detection analyses confirmed that these reflected changes in sensitivity rather than bias. In contrast, inversion disrupted this pattern, attenuating or reversing effects. The results provide strong but qualified support for a hierarchical account of face matching and highlight the boundary conditions of this architecture.

Keywords

unfamiliar faces, face-matching, same-different

Received: 27 August 2024; revised: 1 February 2026; accepted: 6 February 2026

Introduction

People experience considerable difficulties in recognizing and matching photos of unfamiliar faces (Jenkins et al., 2018; Megreya & Burton, 2006). A classic demonstration of suboptimal performance with photos of unfamiliar faces is provided by the *face-matching* task (Bartlett et al., 2024; Bindemann et al., 2013; Megreya & Burton, 2007; White et al., 2014). When presented with two different photos of the same person, participants may perceive them as belonging to two different identities. Similarly, when presented with two images of two different identities, participants may judge them as depicting the same person. These difficulties in "telling faces apart" and "telling faces together" (Burton et al., 2016) arise due to the presence of two sources of variability (White et al., 2022). First, *between-identity* variability, which exists in the appearance of different people, and is the consequence of natural variation in a constrained set of features (eyes, nose, skin tone, etc.). Second, *within-identity* variability in the appearance of the same person, which can result from changes in pose, cosmetics, expression, or other ambient conditions. The two sources of variation have been studied extensively using the *face-matching task* (Megreya & Bindemann, 2018), a methodology that has evolved into an important tool in face research (for a review and a model see Fitousi, 2024).

The present study is set to test fundamental representational and mechanistic aspects of face-matching performance.

[1]Ariel University, Israel

Corresponding Author:
Daniel Fitousi, Ariel University, 3 Kiryat Hamada St., Ariel, Israel.
Email: danielfi@ariel.ac.il

The task deployed here is modeled after the classic *same-different* paradigm (Bamber, 1969; Ben-David & Algom, 2009; Harding & Cousineau, 2022; Krueger, 1978; Proctor, 1981; Ratcliff, 1978; Ratcliff & Hacker, 1981; Walker & Cousineau, 2019)—a well established methodology, that has served psychologists for over 60 years or so, in a broad scope of research questions (for review see Sternberg, 1998). Central phenomena and theories from the vast literature on the same-different task are harnessed here to shed new light on matching photos of unfamiliar faces. The goal is to uncover the processing levels (e.g., physical, nominal; Bruce & Young, 1986; Posner & Mitchell, 1967; Sternberg, 1969) involved in recognizing unfamiliar faces, along with the architectures (serial, parallel; Townsend, 1990) by which these levels are processed, and the stopping rules (self-terminating, exhaustive; Sternberg, 1966) that sustain information accumulation and decision with respect to those representational levels. A related goal is to understand the role of image-variability (Jenkins et al., 2011) in matching unfamiliar faces.

Building Bridges Between Face-Matching and Same-Different Tasks

The face-matching procedure (Megreya & Burton, 2007) bears striking similarities with the well-known same-different task (Bamber, 1969), and as such offers a direct linkage to the canonical same-different task (Bamber, 1969), and its vast attendant literature (Farell, 1985; Krueger, 1978; Proctor, 1981; Sternberg, 1998). In both cases, a pair of stimuli is presented, and the participant's task is that of deciding whether the two are "same" or "different." To expand the theoretical possibilities of the face-matching task and to be able to relate it to the same-different task, the present study added a new condition that is inspired by the seminal study of Posner and Mitchell (1967). These researchers presented participants with pairs of letters (Aa) and asked them to judge these pairs as either "same" or "different." The instructions that were deployed to define "same" were physical identity (AA; physical level), name identity (Aa; nominal level), and rule identity (e.g., both vowels; categorical level). The instructions level was manipulated across blocks of trials, and reaction times (RTs) were recorded. Posner and Mitchell (1967) provided evidence for a hierarchical processing by which physical and then nominal (i.e., name-code) and categorical aspects were perceived one after the other. For example, it took participants more time to respond that the stimuli "AB" are "different" under nominal than physical instructions. Moreover, under nominal instructions, responses to physically identical stimuli (AA) were faster than to nominally identical stimuli (Aa).

This methodology has been successfully imported to the domain of numerical cognition (Dehaene & Akhavein, 1995; Duncan & McFarland, 1980). In these studies, participants were presented with number stimuli (two—8, eight-eight) and asked to indicate whether the stimuli are "same" or "different" on either a physical or nominal level. The question of interest in numerical cognition is whether number magnitude is converted to an a-modal magnitude representation. The results largely supported this hypothesis. RTs in the "different" condition decreased monotonically as a function of the numerical distance separating the numbers. This well-documented distance effect (Moyer & Landauer, 1967) was obtained irrespective of notation and instruction level (physical, nominal), suggesting that it is not bound to surface features of the stimuli or to the demands of the task. The absence of interaction indicates that the effect emerges at a shared processing stage, consistent with the idea that number symbols are automatically converted into an abstract magnitude code. In other words, whether participants compared digits or words, and whether they were instructed to attend to physical or nominal identity, the same underlying representational system produced the distance effect.

The present study applies the classic "same–different" methodology (Farell, 1985; Krueger, 1978; Proctor, 1981; Sternberg, 1998) to photos of unfamiliar faces. Images of facial identities, like letters of the alphabet or number symbols, can be meaningful in various ways. The present study distinguishes between three levels of representation that can and should be dissociated. First, a *featural* level, referring to decisions based on local attributes such as eyes, nose, or mouth. Second, a *physical* level, referring to judgments of whether two images are identical as wholes, regardless of their component features. Third, a *nominal* level, referring narrowly to whether two images depict the same person despite physical variation—analogous to recognizing that "A" and "a" represent the same letter. Importantly, by *nominal*, the present study does not mean broader semantic or social descriptors (e.g., "young male," "friendly"), but specifically the categorical linkage of different images to the same identity. These three levels—featural, physical, and nominal—are conceptually distinct and can be dissociated. To this end, the same–different paradigm is deployed here with images of unfamiliar faces to uncover the underlying mechanisms sustaining these representational levels.

To get a sense of the face-matching task deployed here, the reader is invited to take a look at Figure 1A. Are the two images "same" or "different"? The answer to this question is clear. The two are identical replicas, very much like the stimuli "AA" in Posner and Mitchell (1967) classic study and thus are "same." Now take a look at Figure 1B. Are the two images in this figure "same" or "different"? This becomes tricky because the two images are not physically identical, but they do describe the same person, and are therefore nominally "same," just as the stimuli "Aa" can be viewed as representing the same first letter of the alphabet (Posner & Mitchell, 1967).

In the context of face-matching, the term *nominal* implies that the two faces are same with respect to their

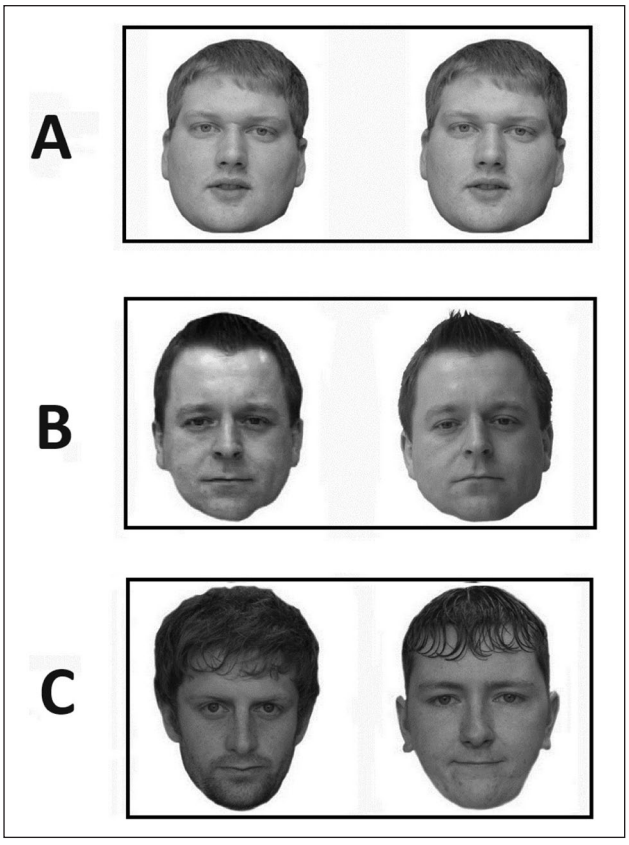


Figure 1. Experiment 1: Examples of the three types of face pairs used in the experiments. Each type of face-pair is characterized by similar relations on the physical (same-different) and nominal (same-different) dimensions. (A) physically same + nominally same, (B) physically different + nominally same, and (C) physically different + nominally different. Note that face-pairs (A) are both physically and nominally "same." Face-pairs (B) are physically "different," but nominally "same," and face-pairs (C) are both physically and nominally "different."

identity, but not necessarily with respect to their physical appearance. Thus, the answer to this question depends on the level of processing adopted by the viewer. If "sameness" is defined at the physical level, then these images are "different"; if, on the other hand, "sameness" is defined at the nominal level, then the two images are "same." These two levels of processing (physical, nominal) are often not probed directly by standard face-matching tasks (but see Davies-Thompson et al., 2009). The reason is that researchers present only variable images of the same identity and define "sameness" only at the nominal level. However, "sameness" relations can be defined at various levels, which may not necessarily converge.

In the present framework, the "nominal" level is defined functionally, as an abstraction across physical and featural variation that supports identity constancy (Ewbank & Andrews, 2008; Ewbank et al., 2008). This characterization differs from the symbolic mapping that links "a" to "A" in classic letter perception studies (Posner & Mitchell, 1967), where the relation is explicitly lexical and conventional. In unfamiliar face matching, no such symbolic convention exists. Instead, the abstraction involves mapping across perceptual variability (e.g., lighting, viewpoint, expression) to a stable identity representation. For this reason, the "nominal" level should be treated not as an exact analog to the letter case, but as a putative abstraction level unique to faces: one that enables generalization without relying on lexical or semantic knowledge (Davies-Thompson et al., 2009). This broader conception allows the hierarchical model to capture how task instructions can direct observers to operate at different levels of representation, while avoiding overextending the original analogy.

Here, the logic of the extended "same-different" paradigm (Posner & Mitchell, 1967) is harnessed to delve deeper into the underlying processing architecture sustaining the matching of unfamiliar identities. A major benefit of the present face-matching paradigm is that it can be linked to the vast empirical and theoretical knowledge accumulated over the last 60 years or so by practitioners of the same-different task (Farell, 1985; Krueger, 1978; Proctor, 1981; Sternberg, 1998). In the next section, a brief review is provided of the central phenomena and ideas in that literature.

Basic Phenomena and Theory in the Same-Different Task

The *fast-same effect* (Bamber, 1969) is a striking phenomenon that has been repeatedly documented in the same-different paradigm (Harding & Cousineau, 2022; Nickerson, 1965; Pachella & Miller, 1976). When sameness is defined at the physical level, participants are faster and more accurate in responding "same" to identical stimuli (AA) than "different" to different stimuli (AE). There have been two popular accounts of this effect. According to the first hypothesis "same" and "different" responses are generated by two different processing mechanisms (Keuss, 1977; Silverman and Goldberg, 1975; Sternberg, 1998). "Same" responses are governed by a rapid, holistic matching process, whereas "different" responses are generated by a slower analytic process. The evidence for this two-processes account arrives from studies showing that distinct factors exhibit dissociated influences on the "same" and "different" responses (Ambler & Proctor, 1976; Egeth & Blecker, 1971).

The two-processes hypothesis has been offered as a solution for the following conundrum. For an efficient comparison of two objects to take place, participants must emit a response as soon as they detect a mismatching attribute. If the pertinent objects are composed of many mismatching features, then participants should stop processing after finding the first mismatch—a strategy known as a *serial self-terminating* (Fitousi, 2021; Sternberg, 1998; Townsend & Ashby, 1983). Evidence for this strategy has been adduced by Bamber

(1969) in the "different" responses. However, according to this logic, if the two objects are identical, participants must compare all features to guarantee that the match is complete and only then emit a "same" response. This predicts longer RTs in "same" than in "different" responses—a pattern that is opposite to the observed fast-same effect (Bamber, 1969). Hence, the solution to this conundrum comes by assuming that "same" and "different" responses are governed by two different architectures (Sternberg, 1998). However, this solution is unsettling because one still has to address a logical obstacle. How do participants know in advance which strategy to apply if "same" and "different" trials typically appear in a random order?

An alternative approach postulates that "same" and "different" responses can be accounted for by a single mechanism (Harding, 2018). According to this hypothesis, differences between "same" and "different" trials occur because of a *rechecking* operation that is performed more often in "different" than "same" trials (Krueger, 1978). This is likely due to internal noise causing more incorrect mismatches in the former than matches in the latter. It is interesting to note that a similar controversy also exists in the face-match literature. Some researchers believe that "same" and "different" face-pairs are governed by two separate mechanisms (Megreya & Burton, 2006), whereas others (Fitousi, 2024) have advocated a single mechanism.

Another important phenomenon in the classics same-different literature has been dubbed the *name-physical disparity effect* (Bamber, 1972). In a typical experiment, sameness is defined at the nominal level, and participants are presented with identical (AA) and nominally identical (Aa) pairs. It is often found that nominally "same" pairs are responded to more slowly than physically identical pairs (Posner & Mitchell, 1967). The common explanation of this effect is that physical matches are based on a physical code, whereas nominal matches are based on a name code (Kroll, 1975; Posner, 1978). Along similar lines, Proctor (1981) argued that a name facilitation generates the effect. When the stimuli are physically identical, participants use only physical analysis; when stimuli are only nominally identical, participants use their name code. However, the role of name code in the same-different task is still under debate (Walker & Cousineau, 2019). The classic study by Posner and Mitchell (1967) suggests that the level of instruction can also affect processing. The same-different literature is rich with empirical phenomena and theoretical accounts. But these two phenomena are the most important ones.

The Same-Different Task with Photos of Unfamiliar Faces

So far, the canonical *fast-same-effect* (Bamber, 1969) and the *name-physical disparity effect* (Bamber, 1972) have not been demonstrated with photos of unfamiliar faces. As noted earlier, the face-matching task was not designed to probe such phenomena. With the present face-matching paradigm at hand, these effects can be studied with unfamiliar faces and uncover the underlying processes. However, it is crucial to mention the allegedly special cognitive aspects of faces that may distinguish them from other objects.

The idea that faces are processed holistically (Farah et al., 1998), and more so than other objects (e.g., letters, words), is a widely accepted notion. However, the term *holistic* itself is fraught with confusion and indeterminacy due to the existence of various possible meanings, definitions, and operationalizations (for a thorough discussion, see Fitousi, 2015; Richler et al., 2012). According to one popular definition (Farah et al., 1998), facial holism refers to "first-order" relations between facial features (e.g., eyes, nose), whereby facial features are dependent on each other, such that the perception of one feature affects the perception of others. According to another influential definition, facial holism refers to "second-order" relations (Tanaka & Farah, 1993), which involve the spatial configuration of features (e.g., the distance between the mouth and nose). To complicate matters, researchers employ different experimental methodologies to measure holism, the construct validity of which has been questioned. Recent studies suggest that these methodologies do not necessarily converge on a single construct (Fitousi, 2015). If facial holism were a unified construct, then hallmark effects—such as the composite face effect (Young et al., 1987), the inversion effect (Yin, 1969), and the part-whole effect (Tanaka & Farah, 1993)—should be strongly correlated. However, Rezlescu et al. (2017) reported weak to nonexistent correlations between these measures.

Moreover, both the "first-order" and "second-order" notions of holism have been subject to criticism (Burton et al., 2015; Fitousi, 2015; Tversky & Krantz, 1969; Wegner & Ingvalson, 2002). For example, Burton et al. (2015) challenged the "second-order" definition based on four main arguments: (a) these theories are not well specified, (b) they cannot accommodate the observation that large changes to configural relations do not impair performance, (c) nor the observation that non-configural changes do impair performance, and (d) face perception appears to be dominated by texture rather than shape. Critiques of the "first-order" account have also been raised (Fitousi, 2015; Sekuler et al., 2004; Tversky & Krantz, 1969; Wegner & Ingvalson, 2002). Note that holistic processing of faces has often been ascribed to parallel or coactive architectures, by which the constituting elements are processed not one-after-the-other, but in tandem (Fifić & Townsend, 2010; Wenger & Townsend, 2005, 2006). Several studies (Cheng & Little, 2025; Cheng et al., 2018; Fitousi, 2021) have adduced evidence against such a holistic strategy, showing that facial features and parts are processed in a serial fashion.

Taken together, these findings may support an alternative—analytic—view of face perception, according to which facial features take precedence over configural information (Fitousi, 2020, 2021). This approach aligns with models of serial processing (Fitousi, 2021). The holistic–analytic debate remains central in the face recognition literature, and these rival perspectives can inspire fruitful models of face processing relevant to our modified face-matching task. Furthermore, integrating concepts from both the same–different literature and traditional face-matching paradigms offers a promising avenue for theoretical development. All of these ideas are considered in relation to the hierarchical model of face matching proposed here.

A Hierarchical Model of Face-Matching

There have been several attempts to characterize the processing system of faces (Burton et al., 1990, 2016). The seminal model by Bruce and Young (1986) is notable for postulating that familiar faces are subjected to hierarchical processing. According to this model, the first processing stage is dedicated to extractions of structural details from the pertinent face, then the observer deciphers a sense of familiarity. At later stages, she extracts semantic information (e.g., occupation), and finally, she recalls the name associated with the face. This hierarchical model has proved successful in explaining recognition with familiar faces both behaviorally and physiologically (Gobbini & Haxby, 2007). But this hierarchical model is inadequate when modeling matching performance with unfamiliar faces, because unfamiliar faces do not have elaborated representations of familiarity, occupation, or name. Moreover, unlike familiar faces, which have image-invariant representation (Bruce & Young, 1986; Haxby et al., 2000), recognition of unfamiliar faces is strongly affected by image-variability (Megreya & Burton, 2006). All of these require a novel framework.

The present work offers a hierarchical model of face-matching. A fundamental hypothesis derived from the model is that unfamiliar faces are subjected to a hierarchical processing strategy along three distinct levels: featural (decisions based on individual parts such as eyes, nose, mouth), physical (decisions about whether two images are identical as wholes), and nominal (judgments of whether two images depict the same person despite physical variation). Thus, featural analysis involves local attributes, physical analysis involves holistic image-level matching, and nominal analysis involves abstract identity recognition. The model predicts that processing of face images in the face-matching task is *serial*, progressing from featural to physical to nominal. Moreover, the model assumes that observers can control the level of processing they operate at, proceeding according to a *self-terminating* rule: participants can halt processing once they have accomplished the intended level of information needed to complete the task. Figure 2 presents the hypothesized stages and their hierarchical organization in this benchmark model.

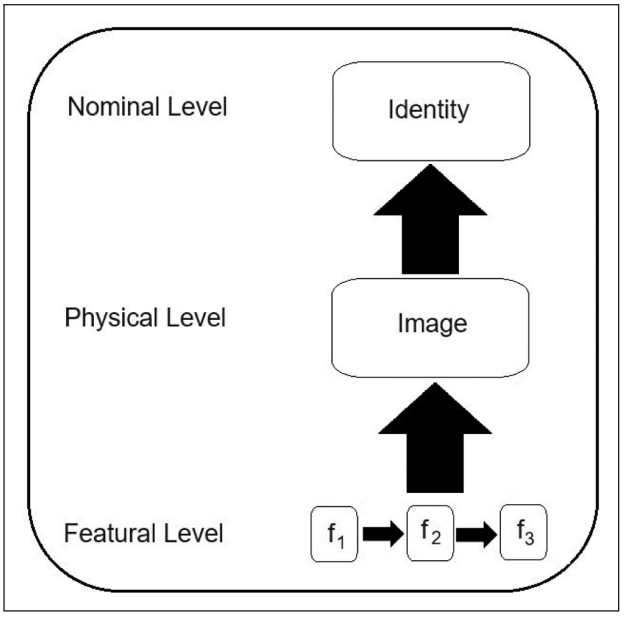


Figure 2. Schematic illustration of the processing stages of unfamiliar faces and their hierarchy. A first feature-matching stage consists of a comparison process with respect to the features depicted by the pertinent images. A second stage is held with respect to the internal representation of the physical images. A third stage occurs at the more abstract level of identity, transcending the image-level representation.

A central role in the present model is ascribed to the intended level of processing—*physical* or *nominal*—as defined by the task instructions (Posner & Mitchell, 1967). Under *physical instructions*, which define sameness at the image level, only featural and physical aspects of the face are considered. Under *nominal instructions*, which define sameness at the identity level, all three levels—featural, physical, and nominal—are engaged.

Consistent with previous models of face matching (Bradshaw & Wallace, 1971; Fitousi, 2024; Megreya & Bindemann, 2018), the matching operation begins at the *featural level*. Here, "features" refer to discrete facial elements such as the eyes, nose, and mouth, as well as low-level visual primitives like edges and contours (Marr, 1982). Evidence suggests that directing attention to this level can improve matching accuracy (Megreya & Bindemann, 2018).

The *physical level* of processing represents the next stage in the hierarchy. It refers to the integration of facial features into a unified, image-based whole. At this stage, the observer has constructed a more holistic representation of the face in the image, but this representation

remains pre-identificational. This level roughly corresponds to the *image-based* stage in other models (e.g., Hancock et al., 2000) and serves as a culmination of earlier featural processing.

Finally, the *nominal level* is reached when the observer activates a conceptual representation of the individual's identity. This process includes the capacity to mentally simulate possible within-person variability in appearance—a key component of real-world face recognition. As Burton et al. (2016) argue, this variability is idiosyncratic and often difficult to learn, posing a challenge for both human observers and models of unfamiliar face processing.

The proposed hierarchical model (see Figure 2) predicts that the matching operation within- and across-levels is held according to a serial self-terminating (i.e., minimum time) architecture (Townsend & Wenger, 2004). *Serial architecture* entails that the featural, physical, and nominal levels are processed one-after-the-other, and not in parallel or in coactivation (Townsend, 1990). *Self-termination* refers to a type of strategy by which processing halts once the extraction of the required information from the intended level (physical) is accomplished (Sternberg, 1966). This is in contrast to an *exhaustive* stopping strategy (Townsend & Nozawa, 1997), by which processing stops only when information from all levels (physical + nominal), or from all features, is extracted, irrespective of the requirements posed by instructions (e.g., physical or nominal).

Evidence for a self-terminating stopping rule at the featural level has been adduced in an earlier study by Bradshaw and Wallace (1971). These authors constructed pairs of faces that varied with respect to the number of shared critical features. Participants were presented with a series of such face-pairs and indicated, per pair, whether the items were "same" or "different." Scanning time (as the overall time taken to respond to all pairs in a block) and error rates increased as a function of the number of critical features. This finding has been interpreted as evidence for a serial self-terminating processing of faces. A recent study by Fitousi (2021) has also documented a serial self-terminating processing with faces in a redundant-target task.

The hierarchical model outlined here (see Figure 2) assumes two types of serial processing. One occurs at the featural level and entails that observers first compare the features in the two faces one-at-a time to arrive at a match. After completion of the featural level, a second serial processing refers to the shift from the physical level to the nominal level. The serial processing at the featural level is associated with another prediction regarding the allegedly holistic nature of faces (Farah et al., 1998). If faces are processed holistically, then features should be matched in parallel. Thus, the holistic approach predicts a *fast-same effect* (Bamber, 1969) under *physical instructions* (Bradshaw & Wallace, 1971; Sternberg, 1998). However, according to the present model, which is consistent with the analytic approach (Fitousi, 2015), processing is held in a serial fashion through all levels. This includes the case in which the observer scans features to confirm or reject physical sameness. Thus, the model predicts the absence of a *fast-same effect* with faces.

The present face-matching task allows for additional tests of the hierarchical model. For example, the serial self-terminating account predicts the emergence of the *name–physical disparity effect* (Bamber, 1972). This is based on the assumption that processing proceeds from the most basic *physical* or image-based aspects (whole-image similarity) to higher *nominal* or identity-based levels (linking different images to the same person). By *nominal*, the model refers specifically to judgments of whether two images depict the same person despite physical variation (analogous to "A" vs. "a" in the seminal study by Posner and Mitchell, 1967), rather than to broader semantic descriptors such as age, gender, or personality traits.

Another set of three Instruction-manipulation tests will also assist us in assessing our model. These tests isolate the effect of *Instruction* while holding stimulus content constant across tasks. First, we tested the instruction effect for "different–different" pairs (DiffPhys_DiffNom), where both tasks involve the same stimulus class and the same overt response ("different"), so any performance change can be attributed to the intended processing level (Posner & Mitchell, 1967). Second, we tested the instruction effect for "same–same" (physically identical) pairs (SamePhys_SameNom), which likewise present identical stimuli across tasks while changing the instructed level (physical vs. nominal), providing a stringent check on whether instruction can re-weight processing when perceptual evidence for sameness is maximal. Third, we tested instruction effects for "nominally same, physically different" pairs (DiffPhys_SameNom), where the *same* stimuli appear under both instruction regimes but require opposite responses ("different" under physical instructions vs. "same" under nominal instructions). Under a hierarchical account, these three comparisons are diagnostic because they dissociate stimulus properties from the instructed representational level, and they therefore allow us to quantify when and how instruction modulates speed and accuracy in face matching.

To complement the RT and error models, we also analyzed performance through the lens of signal detection theory (SDT, Macmillan & Creelman, 2004). The SDT indices provide a compact test of *how* instruction changes performance: a hierarchical account predicts that directing attention to the physical level should primarily improve *discriminability* (higher d'), reflecting more effective use of perceptual information, whereas large shifts in bias (c) would suggest strategic response tendencies rather than a change in the representational level of comparison.

Experiment 1

The experimental paradigm deployed the face-matching task described earlier. It was administered under two types

of instructions (physical and nominal), and with three types of face-pairs: (a) two identical images of the same identity appearing side-by-side, (b) two different images of the same identity appearing side-by-side, and (c) two images of different identities presented one next to the other (see Figure 1). This design is modeled after classic same-different studies by Posner and Mitchell (1967), who used letter stimuli, and by Dehaene and Akhavein (1995), who deployed number stimuli. The face stimuli in the experiment were adapted from the long version of the *Glasgow Face Matching Test* (GFMT; Burton et al., 2010), which is a standard test for unfamiliar face matching.

Methods

Participants. A total of 65 participants took part in the experiment (mean age = 22.40, *SD* = 2.47, 60 females and 5 males). Participants were recruited from Ariel University pool of participants and compensated with a course credit. The study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki. The study was approved by Ariel University Ethical Committee (AU-SOC-DF-20210411). All participants gave their informed consent.

Stimuli and Apparatus. Pairs of faces were constructed from the stimuli in the GFMT (Burton et al., 2010). The face pairs consisted of high-quality grayscale photographs measuring approximately 12×7 cm. The faces conveyed neutral expressions and poses. The experiment included three types of face-pairs (see Figure 1): (a) two physically identical images of a given identity, (b) two different images of the same identity, and (c) two images of different identities. These three types of face-pairs allowed the manipulation of physical and nominal sameness. Thus, face-pairs of type (a) were both physically and nominally "same"; face-pairs of type (b) were physically "different" but nominally "same"; and face-pairs of type (c) were both physically and nominally "different."

Note, however, that the physical and nominal levels were manipulated in a semi-factorial fashion. This is because once the identities of a given face-pair are defined as "different," the physical level is necessarily "different" as well.

Each type of instruction (i.e., physical, nominal) dictated a specific mapping of stimuli to "same" and "different" responses. For example, a face-pair that requires a "different" response under physical instructions might call for a "same" response under nominal instructions. Thus, the goal in constructing the stimuli was to ensure an equal number of "same" and "different" responses under each instruction type, while still presenting all three types of face-pair relationships. To that end, two sets of stimuli, one for each instruction type, were created. Each set consisted of 168 face-pairs, with half (84) designated as "same" and half as "different," according to the relevant instruction.

For the *physical instructions* task, 84 face-pairs were generated that met the criterion for a "same" response under this instruction. This was achieved by taking the original 1 to 84 face-pairs from the GFMT and replacing the right face in each pair with a replica of the left face, resulting in completely identical images. For the "different" stimuli in this task, the first 42 pairs were used from the original GFMT (which depict the same identity but with different images), along with an additional set of 42 face-pairs taken from items 85 to 127 of the GFMT, which depict two different identities. In this way, the number of "same" and "different" trials was kept equal, and the stimuli encompassed all three types of face-pair relationships.

For the *nominal instructions* task, the first 84 items were used from the GFMT that met the criterion for a "different" response under this instruction. These pairs depicted different identities. For the "same" stimuli in this task, items 1 to 42 were deployed from the original GFMT in which the right face in each pair was replaced with a replica of the left face, resulting in face-pairs that were both physically and nominally the same. Additionally, another set of 42 face-pairs corresponding to items 43 to 84 of the original GFMT was used, which depict the same identity but with different images. These were nominally the same but physically different. In this way, the number of "same" and "different" trials was again kept equal, while ensuring that all three types of face-pair relationships were represented.

Identities from the GFMT item set were distributed across both instruction conditions. Items 1 to 84 contributed to both physical- and nominal-instruction blocks, appearing as either identical-image pairs or same-identity/different-image pairs, depending on block requirements. Items 85 to 127 contributed only to the physical-instructions block (different-identity pairs) and did not recur elsewhere. Thus, there was partial overlap of identities across blocks, particularly for items 1 to 84, but no overlap for items 85 to 127. To address this overlap in the statistical analyses, the analysis included random intercepts for items defined at the image level (crossed with task), which captures variation due to both identity and exemplar.

Procedure and Design. The level on which participants based their decision was manipulated across blocks via instructions. For clarity, the terminology adapted here is the one used in Figure 1: (a) same image and same identity, (b) different images of the same identity, and (c) different images of different identities.

Under *physical instructions*, sameness was defined strictly at the image level: only pairs of type (a) were considered "same," whereas pairs of type (b) and (c) were "different." Under *nominal instructions*, sameness was defined at the identity level: both identical pairs and different-image pairs of the same person (types a and b) were judged "same," while type (c) pairs were "different."

Participants were told:

> "On each trial you will be presented with a pair of face images. Your task is to judge whether they are 'same' or 'different' according to a predefined level of similarity that will be given to you at the beginning of each block. In blocks with *physical instructions*, similarity is defined at the image level, and a 'same' response should be given only to identical images. In blocks with *nominal instructions*, similarity is defined at the identity level, and a 'same' response should be given to images of the same person, including identical images."

Each instruction type (physical, nominal) was administered in separate blocks. The order of instructions was counterbalanced across participants: half began with physical instructions and half with nominal instructions. The number of "same" and "different" trials was equated under each instruction type: *Physical instructions*: 84 "same" trials (type a) and 84 "different" trials (42 type b, 42 type c). *Nominal instructions*: 84 "same" trials (42 type a, 42 type b) and 84 "different" trials (type c). In total, each participant completed 338 randomized trials in total.

The experiments were conducted in a quiet laboratory room using Macromedia Authorware software (Macromedia, 1987). Participants were tested individually, and the task took approximately half an hour. On each trial, a face pair was presented at the center of the screen and remained visible until the participant responded. At the beginning of each experimental block, the required level of processing (physical or nominal) was highlighted. Before the experiment began, the experimenter ensured that participants understood the distinction between nominal and physical judgments.

Responses were made using two designated computer keys. Participants were instructed to respond as quickly and as accurately as possible. Both response latency and accuracy were recorded, and the experimenter emphasized the importance of balancing speed and accuracy. To reduce fatigue, participants were given a 2-min mid-way through break after each experimental block of 168 trials.

Results

Analysis Framework and Software. All analyses were conducted in R (R Core Team, 2017). We fit generalized linear mixed-effects models (lme4; Bates et al., 2015) for both outcomes: a Gaussian Generalized Linear Mixed Model (GLMM) with an identity link for RTs (i.e., a linear mixed model) and a binomial GLMM with a logit link for error responses. For RT fixed effects we report Satterthwaite-adjusted tests via lmerTest (Kuznetsova et al., 2017); for errors, we report Wald z-tests. Estimated marginal means (EMMs) and planned contrasts were obtained with emmeans (Lenth, 2023). Note that "errors" denotes incorrect responses (0/1 outcome); thus, RT-fixed effects were evaluated with Satterthwaite-adjusted t-tests in linear mixed models, whereas error fixed effects were evaluated with Wald z-tests in logistic mixed models (binomial logit).

Random-Effects Structure. All models used a fully factorial fixed-effects structure, including the main effects of *instruction* and *condition* and their *instruction* × *condition* interaction. Following current guidance (Barr et al., 2013), we began with the maximal random-effects structure justified by the design: random intercepts and *Instruction* slopes for *participants* and *items* (image-pairs) ([1 + Instruction | participant], [1 + Instruction | item]). Item-level Instruction slopes were included because many items were observed under multiple instruction regimes; item intercepts alone cannot capture item-specific sensitivity to Instruction. Because items are defined as specific image pairs (and therefore belong to a single condition by construction), condition does not vary within an item and an item-level condition slope is not identifiable. If a model yielded a singular fit or failed to converge, we simplified in a pre-specified sequence: (a) remove intercept–slope correlation parameters ([1 + Instruction || . . .]); (b) drop the item-level Instruction slope; (c) drop the participant-level Instruction slope, retaining random intercepts throughout. Inferences reported below are based on the first stable, non-singular model; fixed-effect conclusions were qualitatively unchanged across converged alternatives. Supplementary Materials summarizes the final random-effects specification and diagnostics for each model.

Effect-Size Reporting and Planned Contrasts. For RTs, we report unstandardized regression coefficients B (ms), their standard errors, $\mathrm{SE}(B)$, and associated t-tests. For errors, we report log-odds coefficients B (logit), $\mathrm{SE}(B)$, Wald z-tests, and odds ratios, $OR = \exp(B)$ with 95% confidence intervals (CIs) on the odds scale. For ease of interpretation, we present ORs as values ≥ 1; when $B < 0$ we report $\exp(-B)$ and note this (cf. Erickson et al., 2024). As a descriptive guide, we annotate ORs of ≈ 1.5 (small), ≈ 2.5 (medium), and ≈ 4.5 (large; Sporer & Martschuk, 2014). We tested five a priori, theory-driven contrasts with Bonferroni adjustment across these planned tests.

Sample-Size Strategy, Precision, and Robustness. To harmonize power across experiments and simplify comparison, the primary analyses report the first 40 participants per experiment (equal-N). A combined full-sample analysis statistical conclusions did not differ between equal-N and full-sample fits.

Closed-form power for mixed models is limited; simulation is typically recommended (Green & MacLeod, 2016; Johnson et al., 2015). Our equal-N design (40 participants per experiment; 120 items) yielded narrow confidence intervals for the critical instruction × condition effects, and the pattern of planned contrasts was stable

Table 1. Experiment 1 (Upright; Balanced $N = 40$): Raw By-Subject Means ($\bar{x}$) and Standard Errors. RTs Computed on Correct Trials; Error Rates Computed on All Trials.

Instruction	Condition	RT mean (ms)	RT SE (ms)	Error mean	Error SE
Physical	SamePhys_SameNom	1,069	48.3	0.05	0.01
Physical	DiffPhys_SameNom	1,153	52.4	0.17	0.03
Physical	DiffPhys_DiffNom	1,003	39.2	0.05	0.01
Nominal	SamePhys_SameNom	950	33.3	0.01	0.00
Nominal	DiffPhys_SameNom	1,438	59.8	0.20	0.03
Nominal	DiffPhys_DiffNom	1,440	57.5	0.13	0.02

Note. RT = reaction time; SE = standard error.

when re-fit on the full samples. We therefore emphasize precision (CIs) and convergence of conclusions across equal-N. Traditional ANOVAs largely converge with the GLMM results.

RTs and Error Rates. We next report RTs and error rates to characterize how Instruction and Condition shape performance in this task and to establish the effects that motivate the planned contrasts reported below. We first provide descriptive by-participant means for transparency (Table 1) and then test Instruction, Condition, and their interaction using mixed-effects models for RTs (correct trials) and errors (all trials). Overall, the results are consistent with participants responding to the instructional set, and the instruction × condition interaction suggests that performance depended on the similarity structure across the different pair types.

RTs shorter than 150 ms or longer than 3,800 ms were excluded from analysis. Following common practice and recommendations in the RT literature (see Ratcliff, 1993), the same cutoffs were applied consistently across experiments and conditions. This procedure removed fewer than 5% of trials.

It is important to note that mean RTs computed at the participant level do not necessarily match the EMMs from the mixed model (Lenth, 2016; Luke, 2017; Searle et al., 1980), which can be seen in Figure 3. Whereas participant averages simply collapse across observations, EMMs adjust for participant- and item-level variance as well as potential imbalances in the number of trials per condition. As a result, model-based contrasts provide more accurate estimates of the effects of interest, even if they diverge from raw participant means.

Reaction Times. RTs were modeled with a Gaussian identity-link LMM including fixed effects of instruction (*physical* vs. *nominal*) and condition, plus random intercepts for participants and items and random slopes for Instruction where estimable. Condition used Helmert contrasts with levels L_1 = SamePhys–SameNom, L_2 = DiffPhys–SameNom, L_3 = DiffPhys–DiffNom, and contrasts $C_1 = L_2 - L_1$ ("fast-same") and $C_2 = L_3 - \frac{1}{2}(L_1 + L_2)$. There was a clear Instruction effect: responses were faster under *physical* instructions ($B = -194.97\,\text{ms}$, $SE = 41.21$, $t = -4.73$, $p < .001$; negative B indicates faster under *physical*). Both Condition contrasts were positive, indicating slower RTs for different-image pairs ($C_1 : B = 255.28\,\text{ms}$, $\text{SE} = 16.97$, $t = 15.04$, $p < .001$; $C_2 : B = 86.25\,\text{ms}$, $\text{SE} = 13.46$, $t = 6.41$, $p<.001$). These Condition effects were attenuated under *physical* instructions (negative interactions; $C_1 : B = -383.36\,\text{ms}$, $\text{SE} = 34.55$, $t = -11.10$, $p < .001$; $C_2 : B = -351.13\,\text{ms}$, $\text{SE} = 27.31$, $t = -12.86$, $p < .001$). In sum, identical pairs were fastest, different-image pairs were slower, and adopting a *physical* instruction sped responses overall and reduced the different-image cost.

Errors. There was no reliable overall Instruction difference ($B = 0.02$, $\text{SE} = 0.21$, $z = 0.09$, $p = .931$, OR $= 1.02$, 95% CI $[0.68,1.54]$). For Condition, nominally "same" but different-image pairs (DiffPhys–SameNom) showed much higher error odds than identical pairs ($C_1 : B = 2.39$, $\text{SE} = 0.16$, $z = 14.66$, $p < .001$, OR $= 10.90$, 95% CI $[7.95,14.95]$), whereas the contrast comparing nominally different pairs to the average of the two "same" conditions was not reliable ($C_2 : B = 0.13$, $\text{SE} = 0.12$, $z = 1.04$, $p = .300$, OR $= 1.13$, 95% CI $[0.89,1.43]$). Instruction strongly moderated both contrasts (negative interaction B indicates larger gaps under *nominal* than *physical*; $C_1 : B = -1.67$, $\text{SE} = 0.33$, $z = -5.00$, $p < .001$, OR $= 5.32$, 95% CI $[2.77,10.22]$; $C_2 : B = -1.79$, $\text{SE} = 0.25$, $z = -7.20$, $p < .001$, OR $= 5.96$, 95% CI $[3.67,9.70]$). Overall, identical pairs yielded the fewest errors, nominally same and different-image pairs the most, and the *nominal* instruction accentuated these differences relative to *physical*.

The results suggest that participants were responsive to the instructional set and were capable of shifting their attention to physical and nominal levels by demand. The interaction between instruction and condition entails that participants adopted specific strategies for dealing with the similarity structure inherent in the stimuli across the different types of pairs.

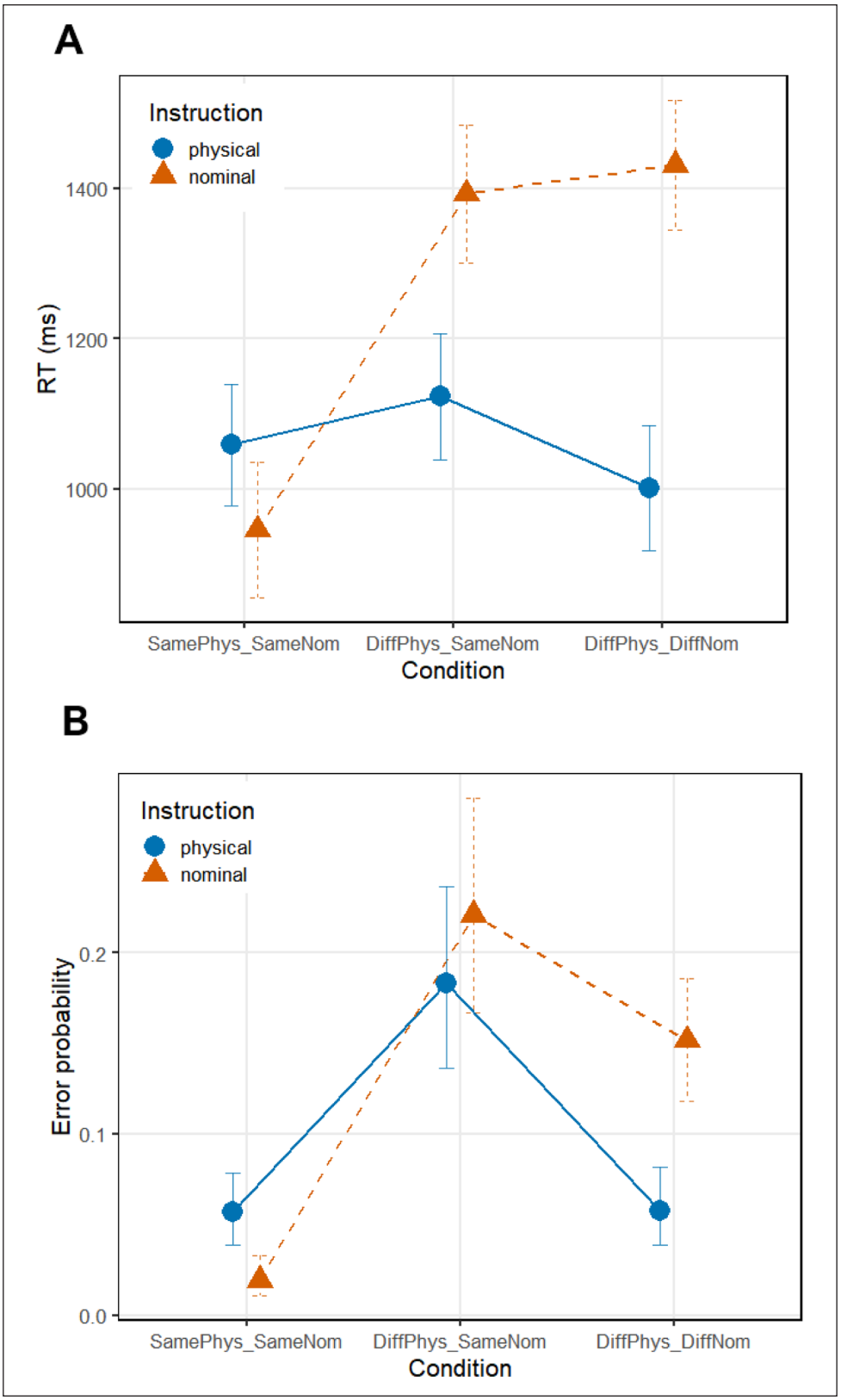


Figure 3. Estimated marginal means by instruction and condition. (A) Reaction time (ms) on correct trials; (B) error probability. Points are model-based EMMs from mixed-effects models Error bars depict 95% model-based CIs on the response scale. Line color/shape denotes Instruction (physical vs. nominal); the *x* -axis shows the three conditions: SamePhys–SameNom, DiffPhys–SameNom, DiffPhys–DiffNom.

Note. EMMs = estimated marginal means; CI = confidence interval.

Planned Contrasts

Fast-Same Effect. First, we tested for the presence of a *fast-same* effect (Bamber, 1969) under *physical* instructions. The effect should emerge as better performance for the "same" condition (same identity + same image) than the "different" condition (different identity + different image). Recall that the hierarchical model predicts the absence of this effect if the processing of facial features is serial. The RTs were in line with the predictions of the hierarchical model, as a *reversed fast-same effect* was obtained: RTs in the "same" condition were longer than RTs in the "different" condition ($B = -57.52$ ms, $\mathrm{SE} = 19.07$, $t = -3.01$, $p = .026$). Errors also conformed with the predictions from our hierarchical model, showing no effect ($B_{\mathrm{logit}} = 0.01$, $\mathrm{SE} = 0.17$, $z = 0.05$, $p = 1.0$, $\mathrm{OR} = 1.01$, 95% CI [0.71,1.43]).

Name–physical disparity (nominal instructions). This effect (Bamber, 1972) entails that under nominal instructions, nominally "same" pairs (same identity + different images) are processed in a serial self-terminating fashion and therefore should yield worse performance than physically identical pairs (see also Proctor, 1981). This prediction was corroborated for RTs ($B = 446.96$ ms, $\mathrm{SE} = 27.79$, $t = 16.08$, $p < .001$) and for error rates ($B_{\mathrm{logit}} = 3.01$, $\mathrm{SE} = 0.29$, $z = 10.32$, $p < .001$, $\mathrm{OR} = 20.41$, 95% CI [11.51,36.19]).

Effect of Instructions on "Different" Pairs. Comparing nominal versus physical instructions when both levels are "different" is a critical test of our model because these conditions involve the same type of stimuli and response, but the intended level of processing changes between the two tasks. Therefore, any difference in performance should be ascribed to the impact of processing level (Posner & Mitchell, 1967). Indeed, performance under physical instructions was faster ($B = 429.06$ ms, $\mathrm{SE} = 44.32$, $t = 9.68$, $p < .001$) and less error-prone ($B_{\mathrm{logit}} = 1.13$, $SE = 0.24$, $z = 4.71$, $p < .001$, $\mathrm{OR} = 3.10$, 95% CI [1.92,4.99]) than under nominal instructions, adducing strong support for our hierarchical processing model.

Effect of Instructions on "Same" (Identical) Pairs. This test compared performance under physical and nominal instructions for physically identical pairs. These are the same images, but the observer's intended level of processing is altered across the two tasks. Our two-stage hierarchical model predicts better performance under physical than nominal instructions. In contrast to this prediction, error rates exhibited a reversed pattern ($B_{\mathrm{logit}} = -1.31$, $SE = 0.31$, $z = -4.14$, $p < .001$, $\mathrm{OR} = 3.70$, 95% CI [1.99,6.88] ; OR reported as $\exp(-B)$ because $B < 0$). RTs showed a similar trend but did not survive Bonferroni correction ($B = -113.75$ ms, $\mathrm{SE} = 44.77$, $t = -2.54$, $p = .110$).

Effect of Instructions on "Nominally Same, Physically Different" Pairs. These stimuli are comparable across the two instruction regimes but require opposite responses ("different" under physical instructions and "same" under nominal instructions). If processing is hierarchical, as our model predicts, then worse performance is expected under nominal than physical instructions. This prediction was supported for RTs ($B = 269.61$ ms, $\mathrm{SE} = 47.11$, $t = 5.72$, $p < .001$), but not for errors ($B_{\mathrm{logit}} = 0.21$, $\mathrm{SE} = 0.24$, $z = 0.86$, $p = 1.0$, $\mathrm{OR} = 1.23$, 95% CI [0.76, 2.01]).

In sum, across the 10 planned contrasts (5 contrast types assessed for both RTs and error rates), 7 supported the hierarchical model, providing strong evidence for the model.

SDT Analysis

The SDT analyses were conducted on trials with RTs between 150 and 3,800 ms. Because SDT assumes a binary decision, a full 2 (instruction) × 3 (condition) analysis is not formally defined in the present design. Accordingly, we focused on the effect of *instruction* by defining signal/noise sets separately for each instruction regime.

Under physical instructions, *signal* trials were defined as physically identical pairs; a *hit* was a "same" response to a signal trial, and a *false alarm* (FA) was a "same" response to a physically different pair. Under nominal instructions, *signal* trials were defined as nominally same-identity pairs; a *hit* was a "same" response to a nominally same-identity pair, and an FA was a "same" response to a nominally different-identity pair. We then computed the canonical equal-variance SDT indices (sensitivity d' and criterion c). Mean d' and c values were computed for each participant under physical and nominal instructions (Figure 4). We tested the effect of *Instruction* using linear mixed-effects models with a fixed effect of Instruction and random intercepts for participants.

Sensitivity (d'). A mixed-effects model revealed a reliable effect of instruction on d' ($t(38) = 4.85$, $p < .001$; equivalently, $F(1,38) = 23.56$, $p < .001$), with higher discriminability under physical than nominal instructions.

Response Bias (c). For criterion c , the effect of instruction was not significant ($t(38) = -1.08$, $p = .285$; equivalently, $F(1,38) = 1.16$, $p = .285$).

Thus, while instructions strongly modulated perceptual sensitivity, they did not yield a reliable shift in decision criterion on the canonical SDT scale.

Discussion

Experiment 1 was designed to evaluate the hierarchical model of face matching (see Figure 2). The task was constructed to dissociate featural, physical, and nominal levels of processing and to test whether performance patterns

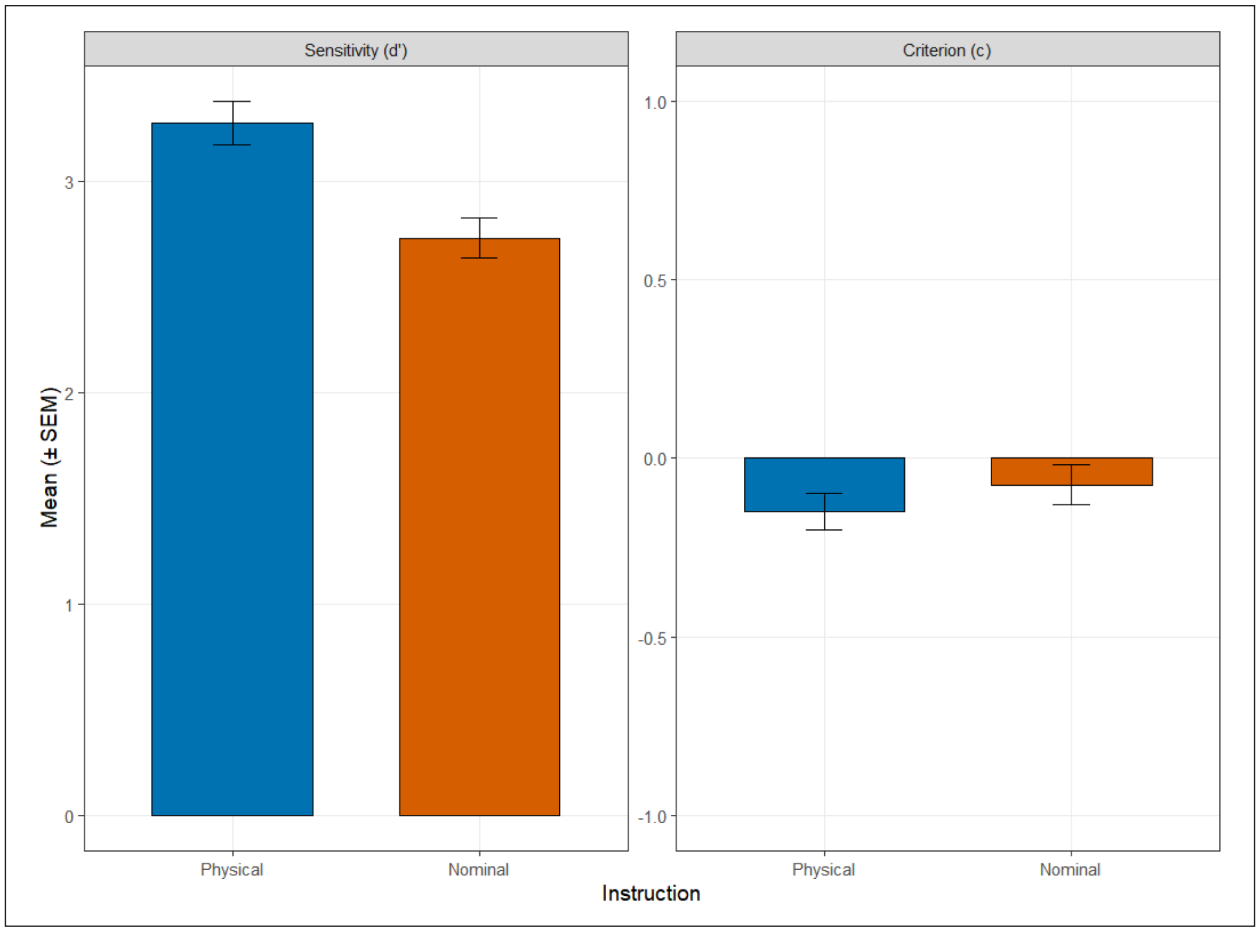


Figure 4. Experiment 1 (upright faces): SDT indices by instruction. Bars show EMMs from linear mixed-effects models computed on participant-level SDT scores, plotted separately for sensitivity (d' ; left) and response bias (c ; right). Error bars denote ± 1 SEM across participants.
Note. EMMs = estimated marginal means; SDT = signal detection theory.

align with a serial, self-terminating architecture. Overall, the results provided solid evidence for the hierarchical model, while also revealing nuances and potential alternative interpretations.

First, the hierarchical model predicts the absence of the *fast-same effect* (Bamber, 1969), which is often taken as a marker of parallel or holistic strategies (Sternberg, 1998). Consistent with this prediction, we did not observe a fast-same advantage. Instead, the pattern was reversed, suggesting a serial, self-terminating mode of processing at the featural level. Second, the *name–physical disparity* effect (Bamber, 1972) emerged, in line with the proposal that nominally "same" pairs require a serial check across images. Together, these findings are compatible with the hierarchical account.

The instruction manipulation further clarified the role of attentional control. Participants appeared able to direct attention to the intended level (physical vs. nominal), and the SDT analyses indicated that instruction effects primarily reflected changes in perceptual sensitivity rather than shifts in decision criterion. This strengthens the claim that instructions modulate the representational level at which similarity is evaluated.

At the same time, one instruction contrast involving identical stimuli (*SamePhys_SameNom*) deviated from the precise predictions of a serial self-terminating architecture, suggesting that even strong top-down control can be constrained by the evidential structure of the stimulus pair. Notably, the instruction manipulation is least diagnostic for *SamePhys_SameNom* (same–same) trials. When the two images are physically identical, evidence for a "same" decision is overwhelming at the earliest perceptual stage, leaving little room for attentional re-weighting of information. Consequently, both instruction sets can be satisfied by the same rapid perceptual match, and any residual instruction effect in this condition is more plausibly attributable to instruction-specific response mapping or verification demands (e.g., confirming identity-level sameness under nominal instructions) than to a substantive change in comparison strategy.

Taken together, the present findings suggest that unfamiliar face matching with upright faces recruits multiple representational levels, that observers can flexibly shift attention between these levels, and that serial self-terminating processing provides a useful—though not exhaustive—description of performance. The hierarchical account, therefore, offers an informative framework, but its predictions should be interpreted alongside similarity- and confusability-based influences.

Experiment 2

One question that remains open is whether the present results generalize to inverted faces (Yin, 1969). The face inversion effect has often been interpreted as evidence for holistic processing of upright faces (Farah et al., 1995). However, several researchers have challenged this conclusion (Fitousi et al., 2021; Sekuler et al., 2004). The evidence presented in Experiment 1 argues against holistic processing (Fitousi, 2015, 2021), as it is mostly consistent with a serial, self-terminating processing architecture. Holistic face processing is typically associated with parallel or coactive architectures that employ an exhaustive stopping rule (Sternberg, 1966). In these strategies, constituent elements are processed simultaneously, and processing continues until all elements have been evaluated (Fifić & Townsend, 2010; Wenger & Townsend, 2005, 2006). This conception of holism is not supported by the results of Experiment 1, which instead highlight a serial, self-terminating strategy more consistent with an analytic mode of processing. It is therefore instructive to examine whether inverting the faces—while preserving the experimental design of Experiment 1—would alter the observed pattern of results.

Methods

Participants. A new group of 40 participants (mean age = 23.1, *SD* = 3.23, 33 females and 7 males), who did not take part in Experiment 1 were recruited from Ariel University pool of participants. An approval of the study was received by Ariel University Ethical Committee (AU-SOC-DF-20210411). All participants filled in an informed consent document.

Stimuli and Design. These were identical to those reported in Experiment 1 except for the fact the face-pairs were presented upside-down. Figure 5 provides examples of the three inverted face-pairs used in Experiment 2.

Results

Table 2 gives actual mean RTs and error rates and their standard errors, as computed from the raw data at the participant level.

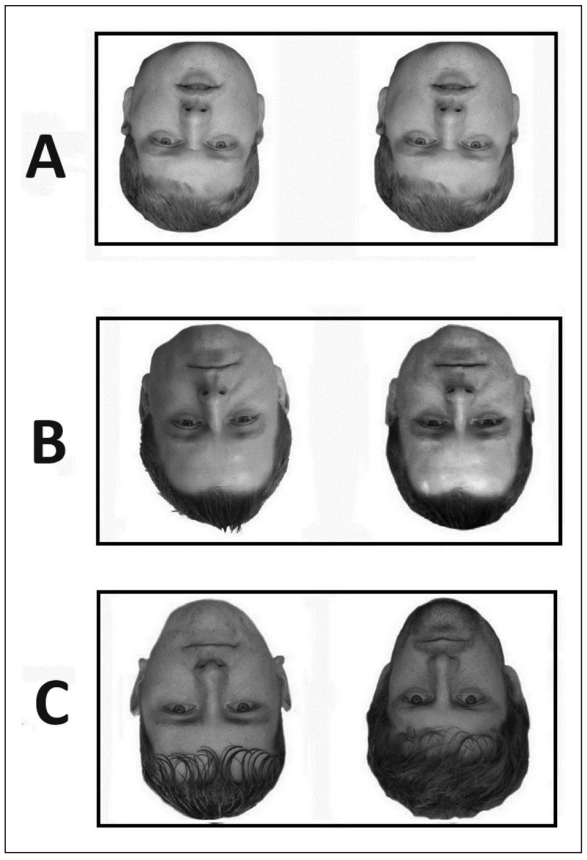


Figure 5. Experiment 2: Examples of the three types of inverted face pairs used in the experiment. Each type of face-pair is characterized by similar relations on the physical (same-different) and nominal (same-different) dimensions. (A) physically same + nominally same, (B) physically different + nominally same, and (C) physically different + nominally different. Note that face-pairs (A) are both physically and nominally "same." Face-pairs (B) are physically "different," but nominally "same," and face-pairs (C) are both physically and nominally "different."

Reaction Times. There was a reliable main effect of condition and an instruction × condition interaction. The overall main effect of Instruction (physical vs. nominal) was not reliable ($B = -133.79\,\text{ms}$, $\text{SE} = 77.15$, $t = -1.73$, $p = .083$; negative B indicates faster RTs under *physical* instructions). Both condition contrasts were positive, indicating slower RTs for different-image pairs ($C_1 : B = 270.38\,\text{ms}$, $\text{SE} = 20.57$, $t = 13.14$, $p < .001$; $C_2 : B = 133.30\,\text{ms}$, $\text{SE} = 16.24$, $t = 8.21$, $p < .001$). The interaction indicated moderation for C_1 but not C_2 ($C_1 : B = -157.14\,\text{ms}$, $\text{SE} = 38.87$, $t = -4.04$, $p < .001$; $C_2 : B = -25.77\,\text{ms}$, $\text{SE} = 31.01$, $t = -0.83$, $p = .406$; Figure 6).

Errors. Instruction (physical vs. nominal) showed no overall change in error odds ($B = 0.02$, $\text{SE} = 0.21$, $z = 0.08$, $p = .934$, OR $= 1.02$, 95% CI [0.68,1.52]). For Condition, errors were strongly elevated for nominally "same" but physically different pairs relative to identical pairs

Table 2. Experiment 2 (Inverted; Balanced $N = 40$): Raw By-Subject Means ($\bar{x}$) and Standard Errors. RTs Computed on Correct Trials; Error Rates Computed on All Trials.

Instruction	Condition	RT mean (ms)	RT SE (ms)	Error mean	Error SE
Physical	SamePhys_SameNom	1,065	54.3	0.04	0.00
Physical	DiffPhys_SameNom	1,300	64.1	0.47	0.04
Physical	DiffPhys_DiffNom	1,319	63.0	0.19	0.03
Nominal	SamePhys_SameNom	1,103	58.2	0.03	0.00
Nominal	DiffPhys_SameNom	1,482	93.2	0.38	0.04
Nominal	DiffPhys_DiffNom	1,459	82.5	0.21	0.03

Note. One cell had $N_{subj} = 39$ (physical–DiffPhys_SameNom) due to missing data. RT = reaction time; SE = standard error.

($C_1 : B = 3.29$, SE $= 0.13$, $z = 24.84$, $p < .001$, OR $= 26.73$, 95% CI $[20.62, 34.63]$), and modestly elevated for physically different pairs relative to the mean of the two "same" cells ($C_2 : B = 0.32$, $SE = 0.10$, $z = 3.24$, $p = .001$, OR $= 1.37$, 95% CI $[1.13, 1.66]$). The Instruction × Condition interaction was not reliable for C_1 ($C_1 : B = 0.37$, $SE = 0.25$, $z = 1.48$, $p = .138$, OR $= 1.44$, 95% CI $[0.89, 2.34]$) but was reliable for C_2 ($C_2 : B = -0.40$, SE $= 0.18$, $z = -2.16$, $p = .031$, OR $= 1.49$, 95% CI $[1.04, 2.13]$; OR reported as $\exp(-B)$ because $B < 0$).

These results suggest that participants were partially responsive to the instructional set with inverted faces, and instructional control was weaker and more selective: instruction modulated RTs (especially for the C_1 contrast) and shifted the distribution of errors (notably reducing the C_2 disparity), while the dominant confusability of nominally-same/physically-different pairs (C_1) largely constrained accuracy regardless of instruction.

Planned Contrasts

To examine the nature of the instruction × condition interaction for inverted faces, we ran the same preregistered, Bonferroni-corrected planned contrasts using LMMs for RTs (Gaussian, identity link) and GLMMs for errors (binomial, logit link). We report regression coefficients B and their standard errors $\mathrm{SE}(B)$; for errors, we also report odds ratios (OR $= \exp(B)$) with 95% CIs. For ease of interpretation, when $B < 0$ we report the reciprocal odds ratio (i.e., $\exp(-B)$), so that reported ORs are ≥ 1; the sign of B indicates direction.

Fast-Same Effect. The *fast-same* effect (Bamber, 1969) with inverted faces was observed for both RTs ($B = 216.32$ ms, SE $= 23.67$, $t = 9.13$, $p < .001$) and errors ($B_{\text{logit}} = 1.78$, $SE = 0.15$, $z = 11.36$, $p < .001$, OR $= 5.93$, 95% CI $[4.36, 8.06]$). This contrasts with Experiment 1, where a reversed fast-same effect was found, undermining the possibility that inverted faces are processed according to our hierarchical model.

Name–Physical Disparity (Nominal Instructions). The effect (Bamber, 1972) was significant for RTs ($B = 348.95$ ms, $SE = 29.43$, $t = 11.85$, $p < .001$) and for error rates ($B_{\text{logit}} = 2.98$, SE $= 0.19$, $z = 14.93$, $p < .001$, OR $= 19.70$, 95% CI $[13.32, 29.13]$). These results conform with the serial self-terminating account.

Effect of Instructions on "Different" Pairs. The effect was not reliable for RTs after Bonferroni correction ($B = 150.97$ ms, SE $= 79.39$, $t = 1.92$, $p = .549$) and was not reliable for error rates ($B_{logit} = 0.23$, SE $= 0.22$, $z = 1.03$, $p = 1.0$, OR $= 1.26$, 95% CI $[0.80, 1.97]$), providing no support for the claim that inverted faces are processed according to our hierarchical model.

Effect of Instructions on "Same" (Identical) Pairs. The effect was not reliable for RTs ($B = 46.63$ ms, SE $= 78.76$, $t = 0.59$, $p = 1.0$) nor for error rates ($B_{\text{logit}} = 0.03$, SE $= 0.26$, $z = 0.14$, $p = 1.0$, OR $= 1.03$, 95% CI $[0.61, 1.75]$), undermining the possibility that inverted faces are processed according to our hierarchical model.

Effect of Instructions on "Nominally Same, Physically Different" Pairs. The effect was not reliable for RTs after Bonferroni correction ($B = 203.77$ ms, SE $= 82.15$, $t = 2.48$, $p = .13$) and was not reliable for errors ($B_{\text{logit}} = -0.32$, SE $= 0.23$, $z = -1.40$, $p = 1.0$, OR $= 1.38$, 95% CI $[0.87, 2.18]$). Thus, instruction did not reliably modulate performance for these pairs.

In sum, across the 10 planned contrasts (5 contrast types assessed for both RTs and error rates), only 2 were consistent with the hierarchical model. These results suggest that, unlike upright faces, inverted faces are less likely to be processed according to the proposed hierarchical architecture.

SDT Analysis: Inverted Faces

Sensitivity (d'). Sensitivity (see Figure 7) was reliably higher under physical than nominal instructions ($B = 0.23$, $SE = 0.06$, $t(38) = 3.89$, $p < .001$), indicating reduced discriminability in the nominal task. EMMs were $d'_{\text{physical}} = 2.41$ (SE $= 0.11$, 95% CI $[2.20, 2.62]$) and $d'_{\text{nominal}} = 1.95$ ($SE = 0.11$, 95% CI $[1.74, 2.16]$).

Bias (c). Instruction also affected the criterion ($B = -0.30$, SE $= 0.05$, $t(38) = -6.55$, $p < .001$),

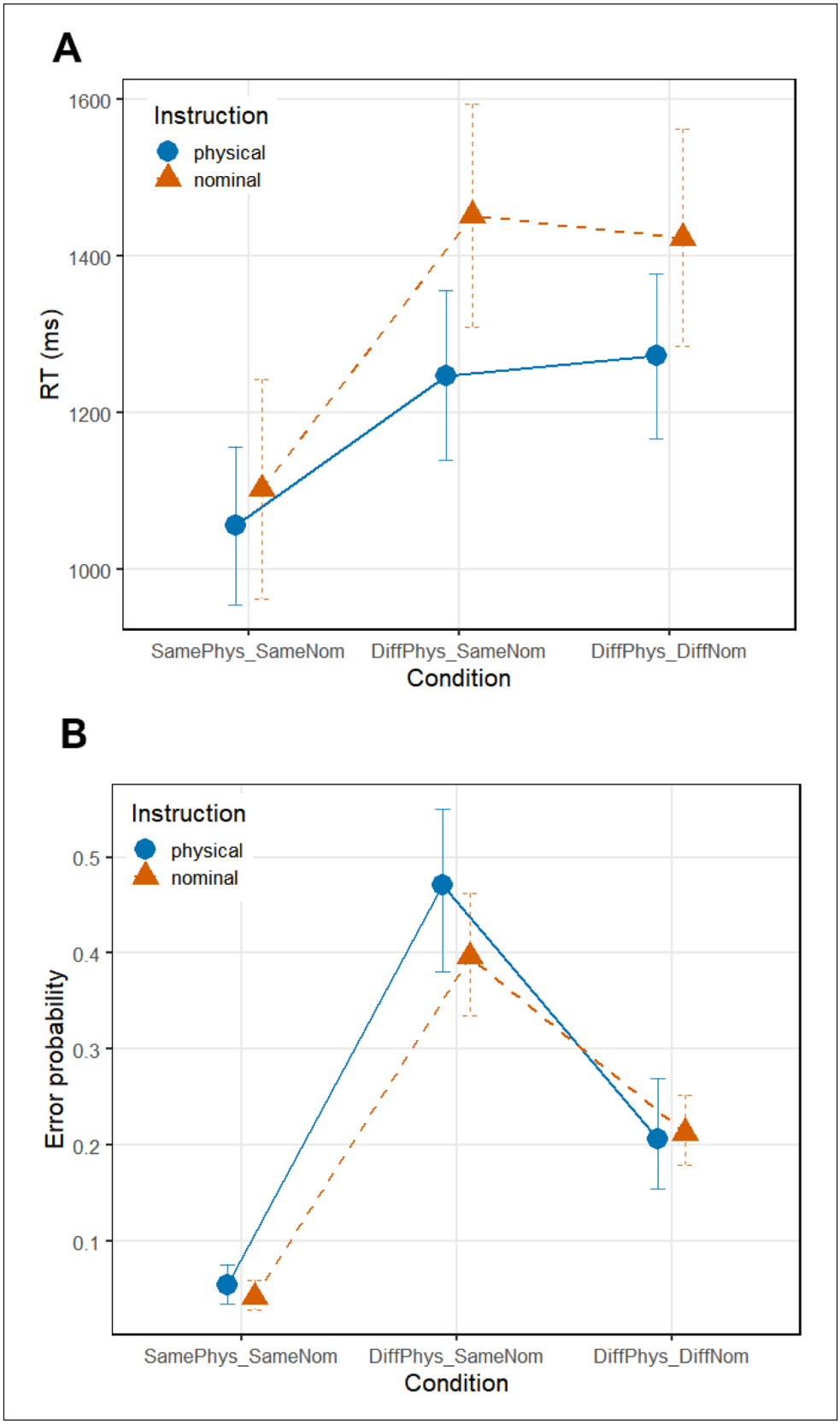


Figure 6. Experiment 2: Inverted faces. Estimated marginal means by instruction and condition. (A) Reaction time (ms) on correct trials; (B) error probability. Points are model-based EMMs from mixed-effects models Error bars depict 95% model-based CIs on the response scale. Line color/shape denotes Instruction (physical vs. nominal); the *x* -axis shows the three conditions: SamePhys–SameNom, DiffPhys–SameNom, DiffPhys–DiffNom.

Note. EMMs = estimated marginal means; CI = confidence interval.

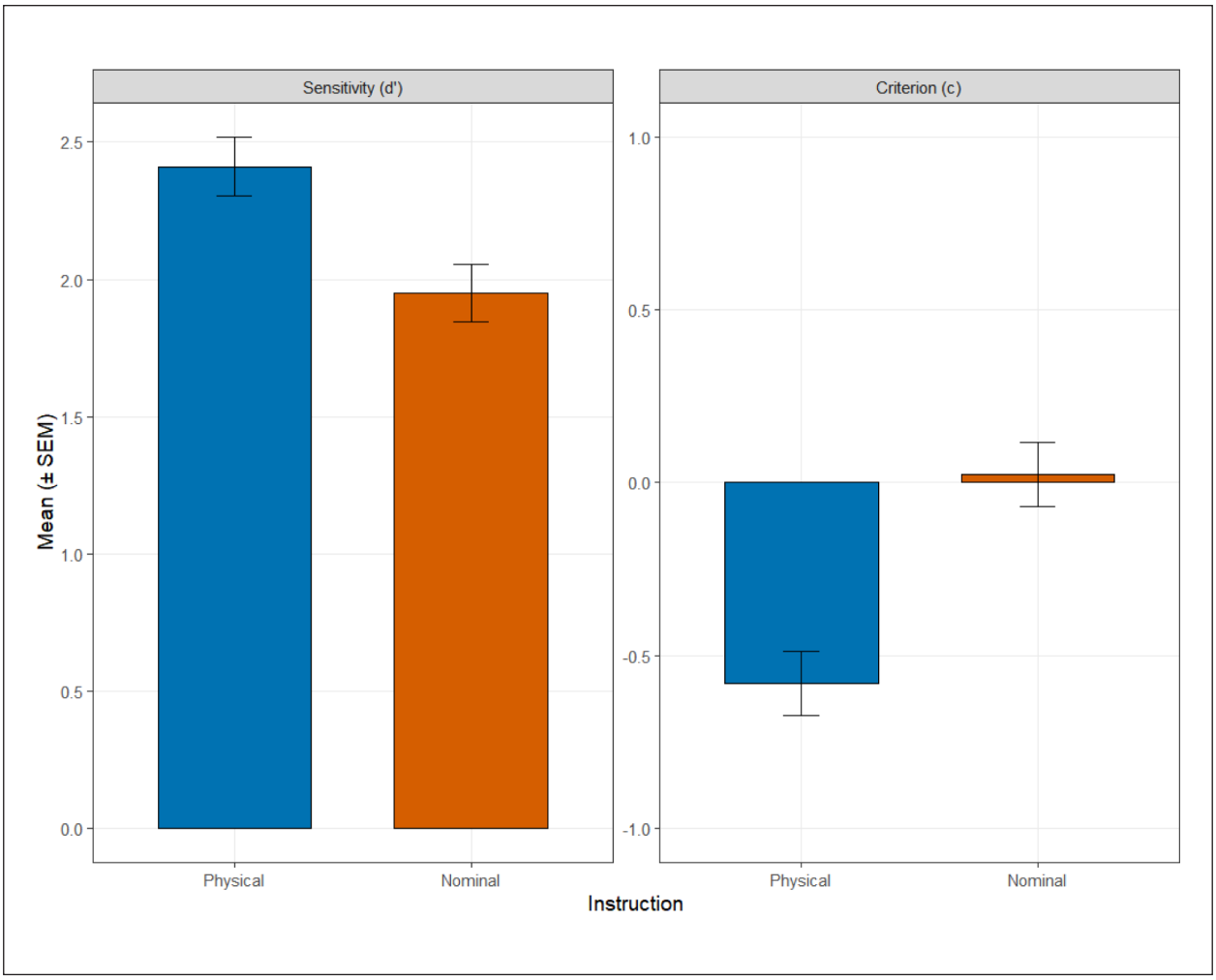


Figure 7. Experiment 2 (inverted faces): SDT indices by instruction. Bars show EMMs from linear mixed-effects models computed on participant-level SDT scores, plotted separately for sensitivity (d' ; left) and response bias (c ; right). Error bars denote $\pm$ 1 SEM across participants.
Note. EMMs = estimated marginal means; SDT = signal detection theory; SEM = standard error of the mean.

reflecting a shift toward a more liberal criterion under physical instructions. The EMMs were $c_{\text{physical}} = -0.58$ ($SE = 0.09$, 95% CI $[-0.76, -0.40]$) and $c_{\text{nominal}} = 0.02$ ($SE = 0.09$, 95% CI $[-0.16, 0.20]$), with the nominal criterion not differing reliably from zero.

Overall, for inverted faces, instructions affected both sensitivity and bias: participants were more sensitive under physical instructions, and they adopted a more liberal criterion in the physical task relative to the nominal task. Recall that for upright faces, instructions affected only discriminability (Figure 7).

Discussion

Experiment 2 tested whether the hierarchical serial account generalizes to face matching with inverted faces. In sum, for inverted faces, participants again responded to the instructional set, indicating that even under degraded conditions, they could shift between physical- and identity-based judgments. However, in contrast to Experiment 1 with upright faces, the predictions of the serial self-terminating model received only limited and inconsistent support. Although some effects were in the expected direction, many planned contrasts were null or reversed, suggesting that inversion substantially weakens the diagnostic pattern predicted by the hierarchical serial account. The SDT finding that instructions also induce a criterion shift supports this conclusion. Taken together, these findings suggest a boundary condition for the hierarchical model: the pattern observed with upright faces is reduced when orientation disrupts normal face processing.

General Discussion

The present study examined whether unfamiliar face matching engages distinct levels of representation (i.e., featural, physical, nominal) and whether these levels operate in a hierarchical, self-terminating fashion.

With upright faces, several key predictions of the hierarchical model were confirmed, including the absence of a fast-same effect and the emergence of the name–physical disparity effect. Additional tests supported the predicted effects of instruction (physical vs. nominal) manipulations, showing that observers could direct attention to different representational levels and process these levels in a predominantly serial manner. SDT analyses further suggested that instructions primarily induced perceptual (sensitivity) rather than decisional (criterion) shifts. One prediction of our model concerning the influence of instructions on the processing of identical stimuli (same identity, same image) was not corroborated, but overall, the evidence for the hierarchical model was solid.

Experiment 2 provided a boundary test of our hierarchical framework. When faces were inverted, several hallmark predictions of the serial model were not obtained. The SDT analyses converged with this pattern, indicating that sensitivity (d') remained higher under physical than nominal instructions, and that instructions also modulated response bias, with a more conservative criterion under nominal than physical instructions. These outcomes suggest that inversion alters the processing architecture, encouraging greater reliance on less efficient heuristics or more undifferentiated similarity judgments.

Taken together, the findings support a qualified version of the hierarchical, self-terminating model. Upright faces appear to promote analytic, stepwise comparisons from features to physical similarity to nominal identity. Inverted faces, however, constrain access to this hierarchy.

Relation to Prior Work on Pictorial Versus Identity Matching

The present findings align with, and contribute to, a growing body of research that distinguishes between multiple levels of face sameness—particularly between pictorial (physical) and identity-level (nominal) representations (Davies-Thompson et al., 2009; Ewbank & Andrews, 2008; Ewbank et al., 2008). Neuroimaging evidence supports the view that unfamiliar face matching relies predominantly on low- to mid-level visual representations rather than abstract, identity-invariant codes. Using fMRI-adaptation, Davies-Thompson et al. (2009) examined whether the fusiform face area (FFA) encodes image-invariant information for familiar and unfamiliar faces. Although repetition suppression was observed for identical images, no adaptation was found across different images of the same unfamiliar individual. In other words, the FFA responded similarly to different images of the same person and to entirely different people, suggesting that representations of unfamiliar faces are image-dependent. This neural evidence complements the hierarchical model advanced here: nominal-level matching for unfamiliar faces is cognitively demanding, error-prone, and closely tied to low-level physical similarity. Only through familiarity do observers appear to develop robust, view-invariant identity codes that support efficient nominal-level matching. The behavioral results reported here reinforce this distinction, revealing strong effects of instructional set and image similarity, consistent with different processing demands at the physical and nominal levels.

In line with this interpretation, the current study's manipulation of sameness at the physical and nominal levels serves as a behavioral analog to the neural dissociations observed in fMRI studies. The observed performance differences across conditions, alongside the shift in implicated processing architecture, support the notion that these forms of face matching are not only psychologically distinct but also neurally dissociable. Future research may combine behavioral paradigms such as the one employed here with neuroimaging techniques in order to directly test the mapping between cognitive architecture and brain activation patterns.

Implications for Image-Variability

Among other important goals, the face-matching task has often been used to study the impact of image variability on face processing (Fitousi, 2024; White et al., 2022). In the present study, performance differences between the physically identical and nominally identical conditions provide a window into the effects of image variability. While the manipulation here involves relatively low-level variability—such as differences introduced by using different cameras or slightly varied poses—it nonetheless represents a controlled form of within-identity variability (White et al., 2022). This manipulation does not capture the full range of naturally occurring within-person variation as observed in studies like Jenkins et al. (2011), but it still allows for a qualitative estimate of the performance cost introduced by image variability under tightly constrained conditions.

Importantly, this study demonstrates that such variability is processed in a serial fashion and that participants can bypass its processing when it is irrelevant to the task. These findings provide a useful foundation for future research examining how different degrees and sources of within-identity variability affect face-matching performance in more ecologically valid settings.

Implications for Models of Face-Matching

A recent paper by Bindemann and Burton (2021) provides comprehensive guidance for the development of future theories of face matching. These authors outline several key factors that any such theory should address. One is *similarity*—because observers have no prior knowledge of the facial identities, they must rely on the perceived degree of similarity between the two images (Fitousi, 2024; Fysh & Bindemann, 2023; Kramer et al., 2025). The present model incorporates several forms of similarity, including featural, physical, and semantic. Another important factor

highlighted by Bindemann and Burton (2021) is *variability*—the same identity may appear across different images due to changes in lighting, pose, and expression. This within-identity variability is contrasted with between-identity variability, which arises from the inherent differences between individuals. The present model offers a fresh perspective on this issue, proposing that both types of variability are governed by indeterminacy in the application of physical and nominal codes. Additional factors discussed by these authors fall under the headings of *decision-making* and *attention*. For instance, they ask whether "match" and "mismatch" decisions are based on a single decision process or on two distinct processes (see Fitousi, 2024), and how attentional strategies influence face-matching performance. The present model directly addresses these questions through its capacity to test the roles of processing architecture (serial vs. parallel), stopping rules (exhaustive vs. self-terminating), and the influence of task instructions and coding strategies on performance in both "match" and "mismatch" trials. Attention is examined here through the lens of visual search strategies and the observer's ability to monitor the level of processing required by the instructions. Both aspects have been central themes in attention research (Posner & Mitchell, 1967).

ORCID iD

Daniel Fitousi https://orcid.org/0000-0003-2369-6680

Ethical Considerations

The study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki. The study was approved by Ariel University Ethical Committee (AU-SOC-DF-20210411). All participants gave their informed consent.

Consent for Publication

Not applicable.

Funding

The author disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was supported by a grant from the Israel Science Foundation (Grant 1498/21) awarded to Daniel Fitousi

Declaration of Conflicting Interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

Data can be downloaded from here https://data.mendeley.com/datasets/zr2dgv59s5/1.

Open Practices

Data or materials for the experiments are available upon request, and none of the experiments was preregistered.

Supplementary Material

The Supplementary Material is available at: qjep.sagepub.com.

References

Ambler, B. A., & Proctor, J. D. (1976). The familiarity effect for single-letter pairs. *Journal of Experimental Psychology: Human Perception and Performance*, *2*(2), 222–234.

Bamber, D. (1969). Reaction times and error rates for "same"-"different" judgments of multidimensional stimull. *Perception & Psychophysics*, *6*, 169–174.

Bamber, D. (1972). Reaction times and error rates for judging nominal identity of letter strings. *Perception & Psychophysics*, *12*(4), 321–326.

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278.

Bartlett, M. L., Carragher, D. J., Hancock, P. J., & McCarley, J. S. (2024). Benchmarking automation-aided performance in a forensic face matching task. *Applied Ergonomics*, *121*, 104364.

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, *67*(1), 1–48.. https://doi.org/10.18637/jss.v067.i01

Ben-David, B. M., & Algom, D. (2009). Species of redundancy in visual target detection. *Journal of Experimental Psychology: Human Perception and Performance*, *35*(4), 958–976.

Bindemann, M., Attard, J., Leach, A., & Johnston, R. A. (2013). The effect of image pixelation on unfamiliar-face matching. *Applied Cognitive Psychology*, *27*(6), 707–717.

Bindemann, M., & Burton, A. M. (2021). Steps towards a cognitive theory of unfamiliar face matching. In M. Bindemann (Ed.), *Forensic face matching: Research and practice* (pp. 38–61). Oxford University Press.

Bradshaw, J. L., & Wallace, G. (1971). Models for the processing and identification of faces. *Perception & Psychophysics*, *9*, 443–448.

Bruce, V., & Young, A. (1986). Understanding face recognition. *British Journal of Psychology*, *77*(3), 305–327.

Burton, A. M., Bruce, V., & Johnston, R. A. (1990). Understanding face recognition with an interactive activation model. *British Journal of Psychology*, *81*(3), 361–380.

Burton, A. M., Kramer, R. S., Ritchie, K. L., & Jenkins, R. (2016). Identity from variation: Representations of faces derived from multiple instances. *Cognitive Science*, *40*(1), 202–223.

Burton, A. M., Schweinberger, S. R., Jenkins, R., & Kaufmann, J. M. (2015). Arguments against a configural processing account of familiar face recognition. *Perspectives on Psychological Science*, *10*(4), 482–496.

Burton, A. M., White, D., & McNeill, A. (2010). The Glasgow face matching test. *Behavior Research Methods*, *42*(1), 286–291.

Cheng, X. J., & Little, D. R. (2025). Second-order facial features are processed analytically in composite faces. *Attention, Perception, & Psychophysics*, *87*(8), 2388–2416.

Cheng, X. J., McCarthy, C. J., Wang, T. S., Palmeri, T. J., & Little, D. R. (2018). Composite faces are not (necessarily) processed coactively: A test using systems factorial technology and logical-rule models. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *44*(6), 833–862.

Davies-Thompson, J., Gouws, A., & Andrews, T. J. (2009). An image-dependent representation of familiar and unfamiliar faces in the human ventral stream. *Neuropsychologia*, *47*(6), 1627–1635.

Dehaene, S., & Akhavein, R. (1995). Attention, automaticity, and levels of representation in number processing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *21*(2), 314–326.

Duncan, E. M., & McFarland, C. E. (1980). Isolating the effects of symbolic distance, and semantic congruity in comparative judgments: An additive-factors analysis. *Memory & Cognition*, *8*, 612–622.

Egeth, H., & Blecker, D. (1971). Differential effects of familiarity on judgments of sameness and difference. *Perception & Psychophysics*, *9*(4), 321–326.

Erickson, W. B., Brown, C., Portch, E., Lampinen, J. M., Marsh, J. E., Fodarella, C., Petkovic, A., Coultas, C., Newby, A., Date, L., Hancock, P. J. B., & Frowd, C. D. (2024). The impact of weapons and unusual objects on the construction of facial composites. *Psychology, Crime & Law*, *30*(3), 207–228.

Ewbank, M. P., & Andrews, T. J. (2008). Differential sensitivity for viewpoint between familiar and unfamiliar faces in human visual cortex. *Neuroimage*, *40*(4), 1857–1870.

Ewbank, M. P., Smith, W. A., Hancock, E. R., & Andrews, T. J. (2008). The m170 reflects a viewpoint-dependent representation for both familiar and unfamiliar faces. *Cerebral Cortex*, *18*(2), 364–370.

Farah, M. J., Tanaka, J. W., & Drain, H. M. (1995). What causes the face inversion effect? *Journal of Experimental Psychology: Human Perception and Performance*, *21*(3), 628–634.

Farah, M. J., Wilson, K. D., Drain, M., & Tanaka, J. N. (1998). What is "special" about face perception? *Psychological Review*, *105*(3), 482–498.

Farell, B. (1985). "same"–"different" judgments: A review of current controversies in perceptual comparisons. *Psychological Bulletin*, *98*(3), 419–456.

Fifić, M., & Townsend, J. T. (2010). Information-processing alternatives to holistic perception: Identifying the mechanisms of secondary-level holism within a categorization paradigm. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *36*(5), 1290–1313.

Fitousi, D. (2015). Composite faces are not processed holistically: Evidence from the Garner and redundant target paradigms. *Attention, Perception, & Psychophysics*, *77*(6), 2037–2060.

Fitousi, D. (2020). Decomposing the composite face effect: Evidence for non-holistic processing based on the ex-Gaussian distribution. *Quarterly Journal of Experimental Psychology*, *73*(6), 819–840.

Fitousi, D. (2021). When two faces are not better than one: Serial limited-capacity processing with redundant-target faces. *Attention, Perception, & Psychophysics*, *83*, 3118–3134.

Fitousi, D. (2024). A signal-detection based confidence-similarity model of face-matching. *Psychological Review*, *131*, 625–663.

Fitousi, D., Rotschild, N., Pnini, C., & Azizi, O. (2021). Understanding the impact of face masks on the processing of facial identity, emotion, age, and gender. *Frontiers in Psychology*, *12*: 743793.

Fysh, M. C., & Bindemann, M. (2023). Understanding face matching. *Quarterly Journal of Experimental Psychology*, *76*(4), 862–880.

Gobbini, M. I., & Haxby, J. V. (2007). Neural systems for recognition of familiar faces. *Neuropsychologia*, *45*(1), 32–41.

Green, P., & MacLeod, C. J. (2016). Simr: An r package for power analysis of generalized linear mixed models by simulation. *Methods in Ecology and Evolution*, *7*(4), 493–498.

Hancock, P. J., Bruce, V., & Burton, A. M. (2000). Recognition of unfamiliar faces. *Trends in Cognitive Sciences*, *4*(9), 330–337.

Harding, B. (2018). *A single process model of the same-different task* [Unpublished doctoral dissertation,. Université d'Ottawa/University of Ottawa].

Harding, B., & Cousineau, D. (2022). Is the fast-same phenomenon that fast? An investigation of identity priming in the same-different task. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *48*(4), 520–546.

Haxby, J. V., Hoffman, E. A., & Gobbini, M. I. (2000). The distributed human neural system for face perception. *Trends in Cognitive Sciences*, *4*(6), 223–233.

Jenkins, R., Dowsett, A., & Burton, A. (2018). How many faces do people know? *Proceedings of the Royal Society B*, *285*(1888), 20181319.

Jenkins, R., White, D., Van Montfort, X., & Burton, A. M. (2011). Variability in photos of the same face. *Cognition*, *121*(3), 313–323.

Johnson, P. C., Barry, S. J., Ferguson, H. M., & Müller, P. (2015). Power analysis for generalized linear mixed models in ecology and evolution. *Methods in Ecology and Evolution*, *6*(2), 133–142.

Keuss, P. (1977). Processing of geometrical dimensions in a binary classification task: Evidence for a dual process model. *Perception & Psychophysics*, *21*, 371–376.

Kramer, R. S., Jones, A., & Fitousi, D. (2025). Face familiarity and similarity: Within- and between-identity representations are altered by learning. *Journal of Experimental Psychology: Human, Perception & Performance*, *51*(7), 927–943.

Kroll, N. E. (1975). Visual short-term memory. In G. H. Bower (Ed.), *Short-term memory* (pp. 153–179). Academic Press.

Krueger, L. E. (1978). A theory of perceptual matching. *Psychological Review*, *85*(4), 278–304.

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). Lmertest package: Tests in linear mixed effects models. *Journal of Statistical Software*, *82*(13), 1–26. https://doi.org/10.18637/jss.v082.i13

Lenth, R. (2016). Least-squares means: the r package lsmeans. *Journal of Statistical Software*, *69*, 1–33.

Lenth, R. (2023). Emmeans: Estimated marginal means, aka least-squares means. *R package version 1.8. 5*.

Luke, S. G. (2017). Evaluating significance in linear mixed-effects models in r. *Behavior Research Methods*, *49*(4), 1494–1502.

Macmillan, N. A., & Creelman, C. D. (2004). *Detection theory: A user's guide*. Psychology Press.

Macromedia. (1987). *Macromedia authorware (version 7)*. Adobe.

Marr, D. (1982). *Vision: A computational investigation into the human representation and processing of visual information*. MIT Press.

Megreya, A. M., & Bindemann, M. (2018). Feature instructions improve face-matching accuracy. *PLoS One*, *13*(3–35), e0193455.

Megreya, A. M., & Burton, A. M. (2006). Unfamiliar faces are not faces: Evidence from a matching task. *Memory & Cognition*, *34*(4), 865–876.

Megreya, A. M., & Burton, A. M. (2007). Hits and false positives in face matching: A familiarity-based dissociation. *Perception & Psychophysics*, *69*(7), 1175–1184.

Moyer, R. S., & Landauer, T. K. (1967). Time required for judgements of numerical inequality. *Nature*, *215*(5109), 1519–1520.

Nickerson, R. S. (1965). Response times for "same"-"different" judgments. *Perceptual and Motor Skills*, *20*(1), 15–18.

Pachella, G. R., & Miller, J. (1976). Stimulus probability and same-different classification. *Perception & Psychophysics*, *19*(1), 29–34.

Posner, M. I. (1978). *Chronometric explorations of mind*. Lawrence Erlbaum.

Posner, M. I., & Mitchell, R. F. (1967). Chronometric analysis of classification. *Psychological Review*, *74*(5), 392–409.

Proctor, R. W. (1981). A unified theory for matching-task phenomena. *Psychological Review*, *88*(4), 291–326.

R Core Team. (2017). *R: A language and environment for statistical computing* [Computer software manual]. https://www.R-project.org/

Ratcliff, R. (1978). A theory of memory retrieval. *Psychological Review*, *85*(2), 59–108.

Ratcliff, R. (1993). Methods for dealing with reaction time outliers. *Psychological Bulletin*, *114*(3), 510–532.

Ratcliff, R., & Hacker, M. J. (1981). Speed and accuracy of same and different responses in perceptual matching. *Perception & Psychophysics*, *30*(3), 303–307.

Rezlescu, C., Susilo, T., Wilmer, J. B., & Caramazza, A. (2017). The inversion, part-whole, and composite effects reflect distinct perceptual mechanisms with varied relationships to face recognition. *Journal of Experimental Psychology: Human Perception and Performance*, *43*(12), 1961.

Richler, J., Palmeri, T. J., & Gauthier, I. (2012). Meanings, mechanisms, and measures of holistic processing. *Frontiers in Psychology*, *3*, 553.

Searle, S. R., Speed, F. M., & Milliken, G. A. (1980). Population marginal means in the linear model: an alternative to least squares means. *The American Statistician*, *34*(4), 216–221.

Sekuler, A. B., Gaspar, C. M., Gold, J. M., & Bennett, P. J. (2004). Inversion leads to quantitative, not qualitative, changes in face processing. *Current Biology*, *14*(5), 391–396.

Silverman, W. P., & Goldberg, S. L. (1975). Further confirmation of same vs. different processing differences. *Perception & Psychophysics*, *17*, 189–193.

Sporer, S. L., & Martschuk, N. (2014). The reliability of eyewitness identifications by the elderly: An evidence-based review. In M. P. Toglia, D. F. Ross, J. Pozzulo, & E. Pica (Eds.), *The elderly eyewitness in court* (pp. 3–37).

Sternberg, S. (1966). High-speed scanning in human memory. *Science*, *153*(3736), 652–654.

Sternberg, S. (1969). The discovery of processing stages: Extensions of Donders' method. *Acta Psychologica*, *30*(0), 276–315.

Sternberg, S. (1998). Inferring mental operations from reaction-time data: How we compare objects. In D. Scarborough & S. Sternberg (Eds.), *An invitation to cognitive science: Methods, models, and conceptual issues* (*Vol. 4*, pp. 365–445). Bradford.

Tanaka, J. W., & Farah, M. J. (1993). Parts and wholes in face recognition. *The Quarterly Journal of Experimental Psychology*, *46*(2), 225–245.

Townsend, J. T. (1990). Serial vs. parallel processing: Sometimes they look like tweedledum and tweedledee but they can (and should) be distinguished. *Psychological Science*, *1*(1), 46–54.

Townsend, J. T., & Ashby, F. G. (1983). *Stochastic modeling of elementary psychological processes*. Cambridge University Press.

Townsend, J. T., & Nozawa, G. (1997). Serial exhaustive models can violate the race model inequality: Implications for architecture and capacity. *Psychological Review*, *104*(3), 595–602.

Townsend, J. T., & Wenger, M. J. (2004). The serial-parallel dilemma: A case study in a linkage of theory and method. *Psychonomic Bulletin & Review*, *11*(3), 391–418.

Tversky, A., & Krantz, D. H. (1969). Similarity of schematic faces: A test of. *Perception & Psychophysics*, *5*(2), 124–128.

Walker, J. A., & Cousineau, D. (2019). Into the mind's eye: Exploring the fast-same effect in the same-different task. *The American Journal of Psychology*, *132*(4), 421–437.

Wegner, M. J., & Ingvalson, E. M. (2002). A decisional component of holistic encoding. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *28*(5), 872–892.

Wenger, M. J., & Townsend, J. T. (2005). *Computational, geometric, and process perspectives on facial cognition: Contexts and challenges*. Psychology Press.

Wenger, M. J., & Townsend, J. T. (2006). On the costs and benefits of faces and words: Process characteristics of feature search in highly meaningful stimuli. *Journal of Experimental Psychology: Human Perception and Performance*, *32*(3), 755–779.

White, D., Kemp, R. I., Jenkins, R., Matheson, M., & Burton, A. M. (2014). Passport officers' errors in face matching. *PLoS One*, *9*(8), e103510.

White, D., Wayne, T., & Varela, V. P. (2022). Partitioning natural face image variability emphasises within-identity over between-identity representation for understanding accurate recognition. *Cognition*, *219*, 104966.

Yin, R. K. (1969). Looking at upside-down faces. *Journal of Experimental Psychology*, *81*(1), 141–145.

Young, A. W., Hellawell, D., & Hay, D. C. (1987). Configurational information in face perception. *Perception*, *16*, 747–759.